

Eeshan Hasan

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RESEARCH STATEMENT

I am fascinated by representational accounts of cognition. I examine whether representations are similar across natural and artificial minds, how we can build representational models of cognition, and how these ideas can inform human–AI systems. I am also interested in characterizing how attentional processes select information to produce task-relevant representations, and in how uncertainty is represented. Beyond individual decision-making, I am interested in understanding differences across individuals and systems, and in how those differences might be leveraged to build systems of collective intelligence. To answer these questions, I use experimental and computational methods.

PROFESSIONAL EXPERIENCE

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|-------------------|--|
| 08/2025 - Current | The Ohio State University
Postdoctoral Researcher
PI: Brandon Turner |
| 05/2017 – 05/2019 | <i>MyCol Healthcare Pvt Limited</i>
Mathematician/Data-Scientist
I built mathematical and predictive models using artificial intelligence and machine learning models to gain insights and improve healthcare systems. |

EDUCATION

- | | |
|-------------------|--|
| 08/2022 – 08/2025 | <i>Department of Psychological and Brain Sciences, Indiana University</i>
<i>Cognitive Science Program, Indiana University</i>
PhD Student (GPA – 4.0/4.0)
Computational Decision-Making Lab. [<i>Advisor: Jennifer Trueblood</i>]
Dissertation Title: The Training-Deployment Cycle of Humans and Machines in Medical Artificial Intelligence |
| 08/2019 – 07/2022 | <i>Department of Psychology, Vanderbilt University</i>
Masters in Psychological Sciences (GPA – 4.0/4.0)
Computational Decision-Making Lab. [<i>Advisor: Jennifer Trueblood</i>]
Transferred to Indiana University along with Jennifer Trueblood |
| 08/2013 – 05/2018 | <i>Department of Mathematics, University of Hyderabad</i>
Integrated Masters in Mathematics (Bachelors + Masters)
<i>First Class with Distinction</i> (GPA – 8.8/10.0) |

AWARDS & HONORS

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| 03/2025 | Outstanding Researcher Award 2025, Cognitive Science Program |
| 06/2024 | Indiana University Bhatt Travel Award |
| 05/2023 | <u>2023 William K Estes Summer Research Award</u> (\$5000) |
| 05/2021 | <u>Computational Modeling Award in Applied Cognition (Cognitive Science Society)</u> |

08/2020	Vanderbilt Psychology Scholarship
08/2019 – 07/2022	Vanderbilt University Graduate Student Scholarship
2018	JRF - Mathematical Sciences (Relinquished)
08/2014-05/2018	<u>KVPY</u>

PUBLICATIONS AND PEER REVIEWED PROCEEDINGS

Hasan, E., Epping, G., Lorenzo-Luaces, L., Bollen, J., & Trueblood, J. S. (2025). One-shot intervention reduces online engagement with distorted content. *PNAS nexus*, 4(3), <https://doi.org/10.1093/pnasnexus/pgaf068>

Hasan, E., Liu, Y., Owens, N.*, Trueblood J. S. (2025) A Registered Report on Presentation Factors that Influence the Attraction Effect *Judgment and Decision Making*. 5 <https://doi.org/10.1017/jdm.2024.27>

Hasan, E., Duhaime, E. P., & Trueblood, J. (2024). Boosting Wisdom of the Crowd for Medical Image Annotation Using Training Performance and Task Features. *Cognitive Research Principles and Implications* <https://doi.org/10.1186/s41235-024-00558-6>

Hasan, E., & Trueblood, J. S. (2024). The Role of Salience in Multialternative Multiattribute Choice. *Proceedings of the Annual Meeting of the Cognitive Science Society* <https://escholarship.org/uc/item/5jq8n5w9>

Hasan, E., Eichbaum, Q., Seegmiller, A. C., Stratton, C., & Trueblood, J. S. (2024). Harnessing the wisdom of the confident crowd in medical image decision-making. *Decision* <https://doi.org/10.1037/dec0000210>

Hasan, E., Eichbaum, Q., Seegmiller, A. C., Stratton, C., & Trueblood, J. S. (2022). Improving Medical Image Decision-Making by Leveraging Metacognitive Processes and Representational Similarity. *Topics in Cognitive Science*. <https://doi.org/10.1111/tops.12588>
[Won the Computational Modeling Prize in Applied Cognition Category from the Cognitive Science Society]

Hasan, E., & Trueblood, J. S. (2022). Representational Smoothing to Improve Medical Image Decision Making. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 44, No. 44). <https://escholarship.org/uc/item/4p6878mm>

*indicates student co-author

MANUSCRIPTS

Hasan E. & Trueblood J. S. (2026) Representational Inconsistency: Understanding When to Re-evaluate Decisions. *Under Review at Psychological Review* https://osf.io/preprints/psyarxiv/9j8tc_v2

Erev, I., Plonsky, O., Roth, Y, **Hasan, E.** (2026) When and which Cognitive Modeling Improves upon the Predictions of Pure Machine Learning Models? The example of decisions from experience *Under Review at Nature Human Behavior*

Wang, Y., **Hasan E.**, Sloutsky, V., Turner, B. (2026) Curse of Simplicity in High-Dimensional Environments *Under Review at npj Science of Learning* <https://www.researchsquare.com/article/rs-10883156/latest>

Hegeman, P. C., **Hasan, E.**, Lee, E., Caplin, A., & Trueblood, J. S. (2026). Miscalibrated AI improves human calibration: An alignment problem for AI-assisted decision education *Under Review at PNAS Nexus*

CONFERENCE POSTERS AND TALKS

Cognitive Psychology

Hasan, E., Sudeep B. Shiffrin, R., Ma, W., Turner B. (2026) Switchboard Frameworks for Model Comparison and Development. *Annual Summer Interdisciplinary Conference 2026*

Hasan, E., & Turner, B. Computational theories of attention in decision making. *Workshop at MathPsych 2026*
Via mathpsych.org/presentation/2079.

Hasan, E., O'Donoghue, E., Broschard, M., Sloutsky, V., Wasserman, E., & Turner, B. (2026, July). Dynamic Selective Attention in Learning, Memory and Decision Making. *Abstract published at MathPsych / ICCM 2026*.
Via mathpsych.org/presentation/2165

Hasan, E., & Trueblood, J. (2025) Using Cognitive Models to Facilitate Human and Artificial Intelligence Collaborative Decision Making. Abstract published at MathPsych / ICCM 2025.
via mathpsych.org/presentation/1833.

Hasan, E., & Trueblood, J. S. (2024). The Role of Salience in Multialternative Multiattribute Choice. *Annual Meeting of the Cognitive Science Society 2024*

Hasan, E., & Trueblood, J. S. (2024). Comparing the impact of medical image classification training on human and machine representations. *Meeting of the Society for Mathematical Psychology 2024*

Hasan, E., Liu, Y., Owens, N., Trueblood J. S. (2024) A Registered Report on presentation factors that influence the attraction effect Judgment and Decision Making. *The Midwest Cognitive Science Society 2024*

Hasan, E., & Trueblood, J. (2023). Computationally Modeling the Role of Bottom-up Attention in Multi-Attribute Choice. *The Annual Meeting of the Psychonomic Society*

Hasan, E., & Trueblood, J. (2023). The Role of Salience-Driven Attention on Multialternative Multiattribute Choice. Abstract published at MathPsych/ICCM/EMPG 2023. Via mathpsych.org/presentation/1203.

Hasan, E., & Trueblood, J. (2022). Representational Smoothing to Improve Medical Image Decision Making. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 44, No. 44).

Hasan, E., & Trueblood, J. (2022). Denoising and Debiasing Medical Image Decisions using Representational Smoothing. *Annual Meeting of the Society of Mathematical Psychology*

Hasan, E., Duhaime, E., Sekhar, T., Trueblood J. S., (2022) Improving Medical Image Classification using Wisdom of the Crowds. *Annual Meeting of the Psychonomic Society*

Hasan, E., Trueblood J.S., Eichbaum, Q., Seegmiller, A., Stratton, C. (2021) Representational Denoising for Improving Medical Image Decision Making *NeurIPS Workshop on Human and Machine Decision Making*
<https://sites.google.com/view/whmd2021/home>

Hasan, E., Trueblood J.S., Eichbaum, Q., Seegmiller, A., Stratton, C. (2021) Improving Medical Image Decision Making by Leveraging Representational Similarity *Society for Mathematical Psychology*
[Accepted as a Talk]

Hasan E. (2021) Learning Multiattribute Choice in an Unknown Environment *Summer Institute on Bounded Rationality at Max Plank Institute*

Hasan, E., Trueblood J.S., Eichbaum, Q., Seegmiller, A., Stratton, C. (2021) Improving Medical Image Decision Making by Similarity Based Aggregation *Psychonomics*

Hasan, E., Trueblood J.S., Eichbaum, Q., Seegmiller, A., Stratton, C. (2021) Improving Medical Image Decision Making by Leveraging Metacognitive Processes and Representational Similarity . *Proceedings of the Cognitive Science Society*

Hasan, E., Trueblood J.S. (2020) Defining a grammar for strategies in Multi-attribute Choice. *Society for Mathematical Psychology.*

Hasan, E., Eichbaum, Q., Daniels, P., Trueblood J.S. (2020) Wisdom of the Crowds for Naturalistic Image Categorization and Decision Making. *Virtual Psychonomics*

Machine Learning

Hasan, E., Trueblood, J., Turner, B., & Nosofsky, R. (2026). *Human-AI alignment in specialized domains where experience is scarce.* 7th International Conference on the Mathematics of Neuroscience and AI, Rome, Italy

Hasan, E. & Trueblood J. (2024) Comparing Novice Perceptual Representations in Medical Domains to Artificial Neural Networks *Midwest Computer Vision Workshop, 2024*

Hasan, E., Duhaime, E. P., & Trueblood, J. (2023). Performance-based Wisdom of the Crowd Algorithms for Medical Image Dataset Labeling *Medical Imaging meets NeurIPS. Workshop at NeurIPS.2024*

Hasan, E. & Trueblood J. (2023) Using Representational Smoothing to Debias and Denoise Medical Decisions. *Artificial Intelligence Open House.*

Hasan E. & Trueblood J. (2021) Representational Denoising to Improve Medical Decisions. *NeurIPS Workshop for Human and Machine Decisions*

Decision Making

Hasan, E., & Trueblood, J. (2022) The Wisdom of the Confident Crowd in Medical Image Decision-making. *Annual Meeting of the Society for Judgment and Decision Making*

Hasan, E., Trueblood J.S., Eichbaum, Q., Seegmiller, A., Stratton, C. (2022) Leveraging Representational Similarity to Improve Medical Image Decision Making *Annual Meeting of the Society for Judgment and Decision Making [Accepted as a Talk]*

Hasan, E., & Trueblood, J. (2023). The Role of Saliency Driven Attention in Multi-Attribute Choice. *The Annual Meeting of the Society for Judgment and Decision Making*

REVIEWING EXPERIENCE

Psychological Review

Npj Digital Medicine

Journal Of Experimental Psychology: Learning Memory and Cognition

Journal of Mathematical Psychology
Judgment and Decision Making
Unireps - Workshop @ NeurIPS
Proceedings of the Cognitive Science Society

TEACHING EXPERIENCE

2023, 2024	The Science of Choice	– <i>Assistant Instructor</i>
2022	Introduction to Statistics	– <i>Assistant Instructor</i>
2020, 2021	Behavioral Decision Making	– <i>Teaching Assistant</i>
2020	Human Sexuality	– <i>Teaching Assistant</i>
2019	General Psychology	– <i>Teaching Assistant</i>

STUDENTS MENTORED

Phillip Hegeman	Computational Decision Making Lab (2023 – 2025)
Ke (Taylor) Lai Honors Student	Computational Decision Making Lab (2021 – 2023)
[Now a PhD Student at Duke]	
Nicole Owens Research Assistant	Computational Decision Making Lab. (2020-2021)
Helena Khalif High School Student	Computational Decision Making Lab. (2020-2021)

SELECTED GRADUATE COURSEWORK

Cognitive Science and Psychology

- Readings at the Interface of Machine Learning and Cognitive Science
- Philosophy of Cognitive Science
- Computational Models of Attention
- Models of Human Memory
- Computational Neuroscience of Vision
- Computation Cognitive Modeling
- Computational Neuroscience
- Bayesian Cognitive Modeling
- Emotions
- Scientific Writing
- Psychological Measurement
- Teaching of Psychology
- Human Cognition
- Developmental Psychology
- Introduction to Cognitive Science

Mathematical Sciences

- Algebra – I, II, III (Group Theory, Ring Theory, Field Theory)
- Real Analysis – I, II, III (Real, Multivariate)
- Complex Analysis
- Functional Analysis and Advanced Functional Analysis
- Measure Theory and Integration
- Representation Theory
- Dynamical Systems
- Commutative Algebra
- Game Theory

Philosophy

- Philosophy for Cognitive Science
- Science and Values

- Western Philosophy III (Hume, Berkeley and Locke) Empiricism

REFERENCES

Jennifer Trueblood (jstruebl@iu.edu)

Ruth N. Halls Professor, Psychological and Brain Sciences
Director, Cognitive Science Program
Indiana University Bloomington

Johan Bollen (jbollen@iu.edu)

Chair, of Informatics and Cognitive Science
Luddy School of Informatics, Computing, and Engineering
Cognitive Science Program
Indiana University Bloomington

Robert Goldstone (rgoldsto@iu.edu)

Distinguished Professor and Chancellor's Professor, Psychological and Brain Sciences
Fellow, American Academy of Arts and Sciences
Indiana University Bloomington

Robert Nosofsky (nosofsky@iu.edu)

Distinguished Professor and Chancellor's Professor, Psychological and Brain Sciences
Fellow, American Academy of Arts and Sciences
Indiana University Bloomington

Brandon M. Turner (turner.826@gmail.com)

Professor, Psychological Sciences
Ohio State University